

1. Build Array from Permutation

Easy

Topics

Companies

Hint

Given a **zero-based permutation** `nums` (**0-indexed**), build an array `ans` of the **same length** where `ans[i] = nums[nums[i]]` for each $0 \leq i < \text{nums.length}$ and return it.

A **zero-based permutation** `nums` is an array of **distinct** integers from 0 to `nums.length - 1` (**inclusive**).

Example 1:

Input: `nums = [0,2,1,5,3,4]`

Output: `[0,1,2,4,5,3]`

Explanation: The array `ans` is built as follows:

```
ans = [nums[nums[0]], nums[nums[1]], nums[nums[2]], nums[nums[3]], nums[nums[4]], nums[nums[5]]]  
    = [nums[0], nums[2], nums[1], nums[5], nums[3], nums[4]]  
    = [0,1,2,4,5,3]
```

Example 2:

Input: `nums = [5,0,1,2,3,4]`

Output: `[4,5,0,1,2,3]`

Explanation: The array `ans` is built as follows:

```
ans = [nums[nums[0]], nums[nums[1]], nums[nums[2]], nums[nums[3]], nums[nums[4]], nums[nums[5]]]  
    = [nums[5], nums[0], nums[1], nums[2], nums[3], nums[4]]  
    = [4,5,0,1,2,3]
```

Constraints:

`1 <= nums.length <= 1000`

`0 <= nums[i] < nums.length`

The elements in `nums` are **distinct**.

Follow-up: Can you solve it without using an extra space (i.e., $O(1)$ memory)?

PROGRAM

```
class Solution {  
  
    public int[] buildArray(int[] nums) {  
  
        int[] ans = new int[nums.length];  
  
        for (int i = 0; i < nums.length; i++) {
```

```
ans[i] = nums[nums[i]];
```

```
}
```

```
return ans;
```

```
}
```

```
}
```

OUTPUT

The screenshot displays the LeetCode interface for the problem "Build Array from Permutation". The submission status is "Accepted" with 140/140 testcases passed. The code is written in Java and is shown in the "Code" tab. The "Testcase" tab shows the input and output for the test cases.

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Accepted 140 / 140 testcases passed
VTU30221 submitted at Sep 12, 2026 19:13

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Runtime
1 ms | Beats 99.56%
Memory
47.10 MB | Beats 52.37%

Code | Java

```
1 class Solution {  
2     public int[] buildArray(int[] nums) {  
3         int[] ans = new int[nums.length];  
4         for (int i = 0; i < nums.length; i++) {  
5             ans[i] = nums[nums[i]];  
6         }  
7         return ans;  
8     }  
9 }
```

Testcase | **Test Result**

Accepted Runtime: 0 ms

Case 1 Case 2

Input

nums =
[0,2,1,5,3,4]

Output

[0,1,2,4,5,3]

Expected